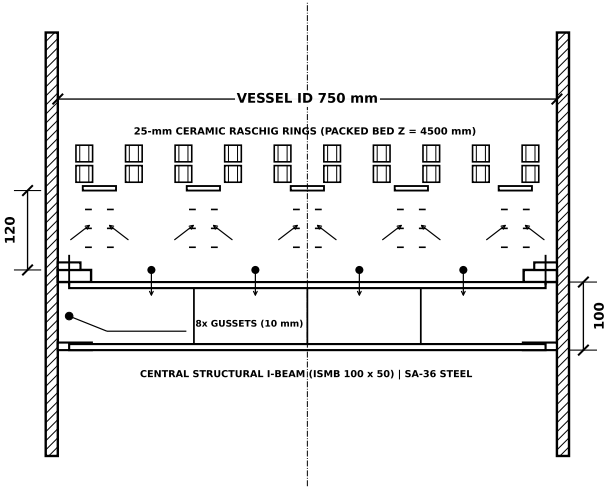
VIEW 1 — PACKING SUPPORT GRID (ELEVATION)

GAS-INJECTION MULTI-BEAM TYPE (>85% FREE AREA) — SCALE 1:20



VIEW 2 — SUPPORT GRID (TOP PLAN VIEW)

SEGMENTAL PANELS & STRUCTURAL BEARING — SCALE 1:20

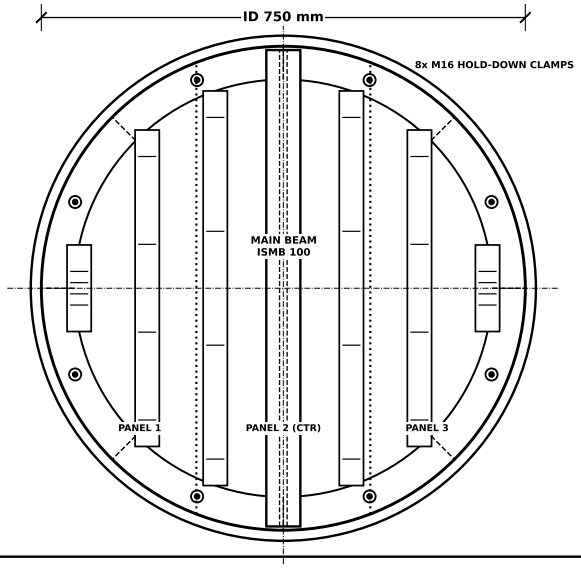
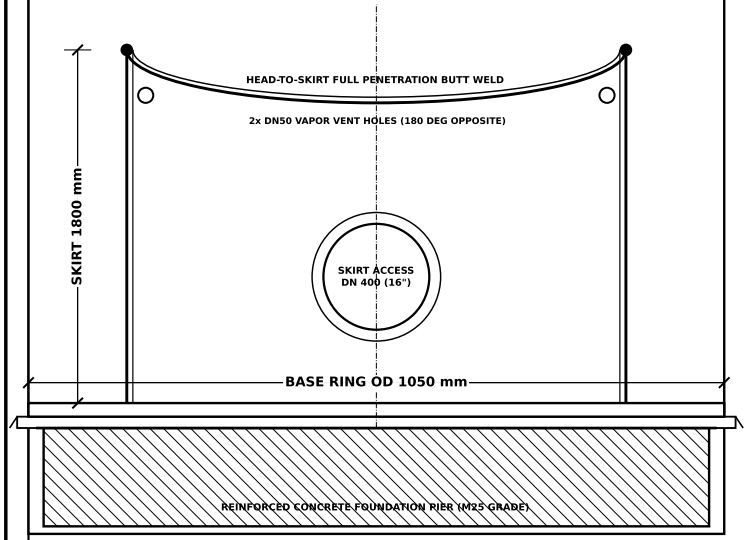


TABLE 1 — PRELIMINARY STRUCTURAL DESIGN BASIS

DESIGN PARAMETER	VALUE	SPECIFICATION / BASIS
Operating Grid Load	16.08 kN (1.64 t)	Dry pack + liq holdup + tare
Flooded Upset Load	26.39 kN (2.69 t)	100% void flooded emergency
Basic Wind Speed (Vb)	160 km/h (44.4 m/s)	**IS 875 (Part 3) / Cat. 2**
Wind Base Moment	18.52 kN*m	Calculated at skirt base ring
Seismic Design Zone	Zone III (Z = 0.16)	IS 1893 (Part 4) / I = 1.5
Central Support Beam	ISMB 100 x 50	SA-36 Steel (Zx = 25.7 cm3)
Max Beam Deflection	0.32 mm (<0.75 mm)	delta <= L/1000 allowable
Foundation Anchorage	8x M24 SA-193 B7	950 mm BCD (Tension = 14.8 kN)
Skirt Shell & Base Plt	SA-36 (10 & 20 mm)	Max comp. stress 12.4 MPa
Foundation Bearing	1.85 MPa (<6.25)	M25 Concrete (Allowable 6.25)

VIEW 3 — CYLINDRICAL SKIRT & BASE RING

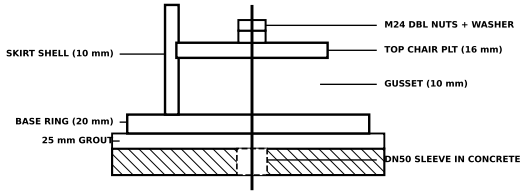
ANCHOR BOLTS & FOUNDATION CHAIRS — SCALE 1:20



VIEW 4 — STRUCTURAL FABRICATION & CONNECTION DETAILS

DETAIL A: ANCHOR CHAIR | DETAIL B: SUPPORT LEDGE — SCALE 1:5

DETAIL A — ANCHOR BOLT CHAIR & FOUNDATION CONNECTION



DETAIL B — SUPPORT LEDGE RING & BEAM END BEARING

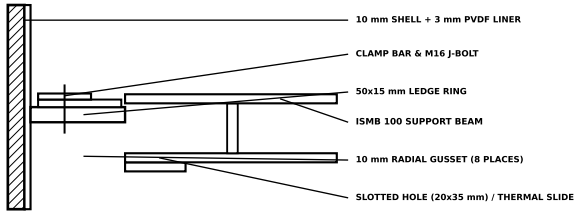


TABLE 2 — BILL OF MATERIALS (BOM) — STRUCTURAL

MARK	STRUCTURAL ITEM	MATERIAL	DIMENSIONS / SECTION	QTY
S1	Gas-Injection Support Grid	FRP / Ti Gr 2	740 mm Dia (Slotted Chimney)	1 SET
S2	Support Ledge Ring	SA-516 Gr 70	**750 OD x 650 ID x 15 mm**	1 PC
S3	Ledge Radial Gussets	SA-516 Gr 70	100 x 50 x 10 mm Plate	8 PCS
S4	Central Support Beam	SA-36 Steel	ISMB 100 x 50 (L = 740 mm)	1 PC
S5	Beam Bearing Cleats	SA-36 Steel	L 75 x 75 x 8 x 100 mm	2 PCS
S6	Cylindrical Skirt Shell	SA-36 Steel	**750 OD x 10 mm x 1800 mm**	1 PC
S7	Annular Base Ring	SA-36 Steel	1050 OD x 730 ID x 20 mm	1 PC
S8	Anchor Chair Gussets & Top	SA-36 Steel	10 mm Gussets / 16 mm Top	8 SETS
S9	Foundation Anchor Bolts	SA-193 B7	M24 x 600 mm Studs (Dbl Nut)	8 PCS

PRELIMINARY STRUCTURAL FABRICATION & ERECTION NOTES

1. **PRELIMINARY BASIS: STRUCTURAL SIZING & LOADS ARE ACADEMIC PRELIMINARY VALUES AND REQUIRE FINAL MECHANICAL & STRUCTURAL VERIFICATION.**
2. **WELDING: PER ASME SEC IX & AWS D1.1. SKIRT-TO-HEAD IS FULL-PENETRATION** BUTT WELD WITH 100% RADIOGRAPHY (ASME UW-11 CAT B).
3. ANCHORAGE: 8x M24 (SA-193 B7) BOLTS TORQUED TO 450 N*m IN STAR PATTERN.
4. GROUT: 25 mm NON-SHRINK GROUT (MIN COMPRESSIVE STRENGTH 50 MPa AT 28 DAYS).
5. EXPANSION: SLOTTED HOLE (20x35 mm) AT BEAM BEARING PERMITS THERMAL SLIDE.
6. SKIRT VENTS: 2x DN50 VENT HOLES AT 180 DEG PREVENT HAZARDOUS GAS POCKETS.

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
PACKED-BED ABSORBER (C-101) — PACKING SUPPORT & SKIRT BASE CAD	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	1:20 METRIC
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	PEDD-PBA-03	29 / 09 / 2026

PACKED ABSORBER (C-101) | 0.25 m3/s GAS (2% Br2) | WATER: 17.25 kg/s | ASME SEC VIII DIV 1 / B16.5